

Body-fitted finite-volume NACA0012 verification

Actual body-fitted Gmsh meshes; strict physical relative error <3%.

Scheme: {'momentum_convection': 'linear-upwind', 'pressure_gradient': 'gauss-linear', 'face_interpolation': 'projected-linear', 'wall_gradie

CPU calculations; no new GPU or matched TensorLBM result.

Positive mesh geometry and nonorthogonality gate precede all solves.

scale-2: 3934 cells; converged=True; accepted=False

{'drag': 0.12756952528780408, 'lift': 0.23207023502685248}

{'drag_relative_error': 0.0213198, 'lift_relative_error': 0.1553623}

scale-1: 15018 cells; converged=True; accepted=False

{'drag': 0.1271412456582323, 'lift': 0.21632053381166771}

{'drag_relative_error': 0.017891, 'lift_relative_error': 0.0769524}

Full independently replayed equations: True

Refinement sequence accepted: False

Asymptotic grid regime qualified: False

Reference paper uses open top/bottom boundaries.

Current diagnostic imposes freestream there; matching remains unresolved.

Problem, numerical method and reproducibility

Body-fitted finite-volume NACA0012 verification

Problem and reference

Steady incompressible laminar NACA0012 flow at $Re=1000$ and incidence 4 degrees, unit chord, domain 20 by 16 m and leading edge (6,8) m. The freestream vector is $(\cos(4 \text{ deg}), \sin(4 \text{ deg}))$ m/s. The body is no-slip; prescribed velocity is applied at the inlet and top/bottom far-field boundaries, and outlet pressure is zero. Drag/lift use aerodynamic axes. Comparisons use vector-digitized present-study markers in [Di Ilio et al. (2020), figures 10 and 11](<https://arxiv.org/html/2006.10487>), with ± 0.0001 graph-reading intervals and worst endpoint relative errors. The paper specifies open top/bottom boundaries, whereas this diagnostic retains prescribed freestream there. The paper also uses a 36C by 16C domain and places the quarter-chord point 12C from the inlet. Both boundary and domain differences must be resolved before claiming a matched reference benchmark. No matched 4-degree tabulated C_p reference is available.

Mesh and discretization

Gmsh 4.15.2 frontal Delaunay generates conforming triangles with physical boundary tags; cylinder CAD arcs or a closed NACA0012 interpolating spline define the body. Physical wall faces are linear chords. See the [Gmsh official manual](<https://gmsh.info/doc/texinfo/gmsh.html>) for algorithm 6 and Distance/Threshold fields. Neither Cartesian obstacle masks nor staircase boundaries are used. Positive volumes and projected face distances and a maximum nonorthogonality below 70 degrees are required before solving. The 70-degree project gate is a screening criterion, not proof of sufficient accuracy.

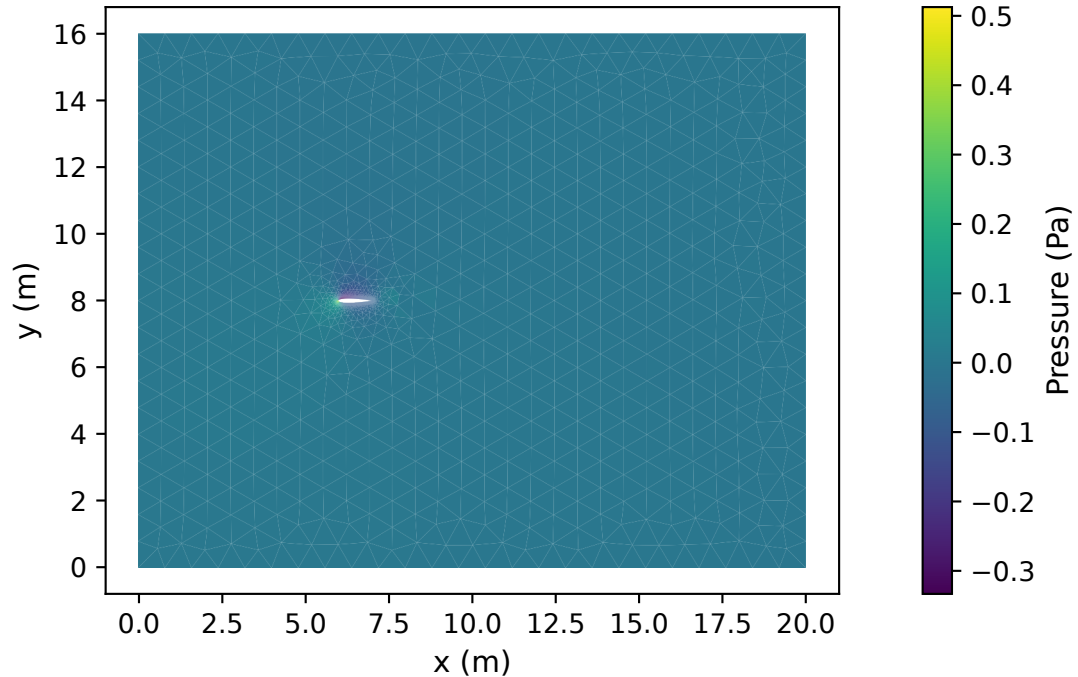
The recorded discretization is ``{"face_interpolation": "projected-linear", "momentum_convection": "linear-upwind", "pressure_gradient": "gauss-linear", "wall_gradient": "displacement-correction"}``. Linear upwind uses the least-squares gradient of the upstream cell and the actual face-to-cell displacement; its conservative deferred correction is inspired by [OpenFOAM linearUpwind](<https://github.com/OpenFOAM/OpenFOAM-dev/blob/master/src/finiteVolume/interpolation/surfaceInterpolation/schemes/linearUpwind/linearUpwind.C>). This is unlimited reconstruction, so no bounded/TVD claim is made. Gauss pressure gradients use geometric face interpolation and the same boundary face pressure used by pressure traction. Nonorthogonal diffusion, physical Rhie-Chow flux and no-slip reconstructed molecular stress are retained. Cylinder acceleration uses a full-grid coupled Picard matrix with an independent response check; NACA uses SIMPLE and Anderson mixing. CPU sparse solves check true residuals. No reference value enters the equations, initialization or relaxation.

Reproduction and acceptance

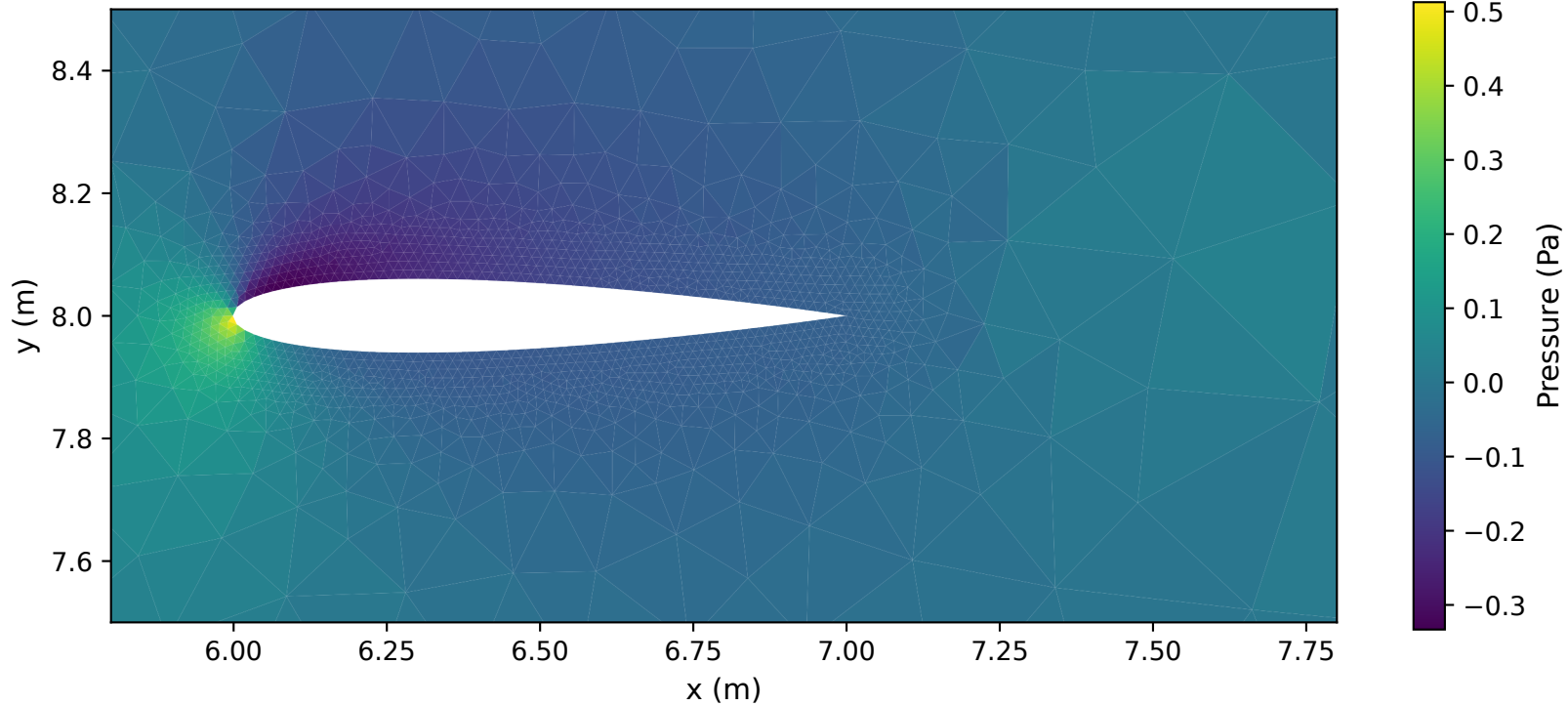
```
```bash
uv pip install --python /path/to/python gmsh numpy scipy matplotlib
OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 PYTHONPATH=src python -m
tensorfvm.verification.external_gmsh --case naca --scales 2.0 1.0 --solver conservative-linear-
upwind --max-iterations 1800 --output results/naca
PYTHONPATH=src python scripts/audit_external_equations.py results/naca
PYTHONPATH=src python -m tensorfvm.verification.external_mesh_report results/naca
```
```

Raw mesh/field files, velocity/pressure arrays, face mass fluxes, separate tractions, full histories, linear residuals and SHA256 source/artifact manifests accompany this report. Each physical relative error must be strictly below 3%; steady equation residuals must be below $1e-6$ and the physical flux fixed-point defect below $1e-7$. Source-preserving independent NumPy replay checks full configured equations. Three runs do not establish an asymptotic grid regime; qualification of the finest run does not qualify the entire refinement sequence. GPU and matched TensorLBM runs have not been performed for this new unstructured path.

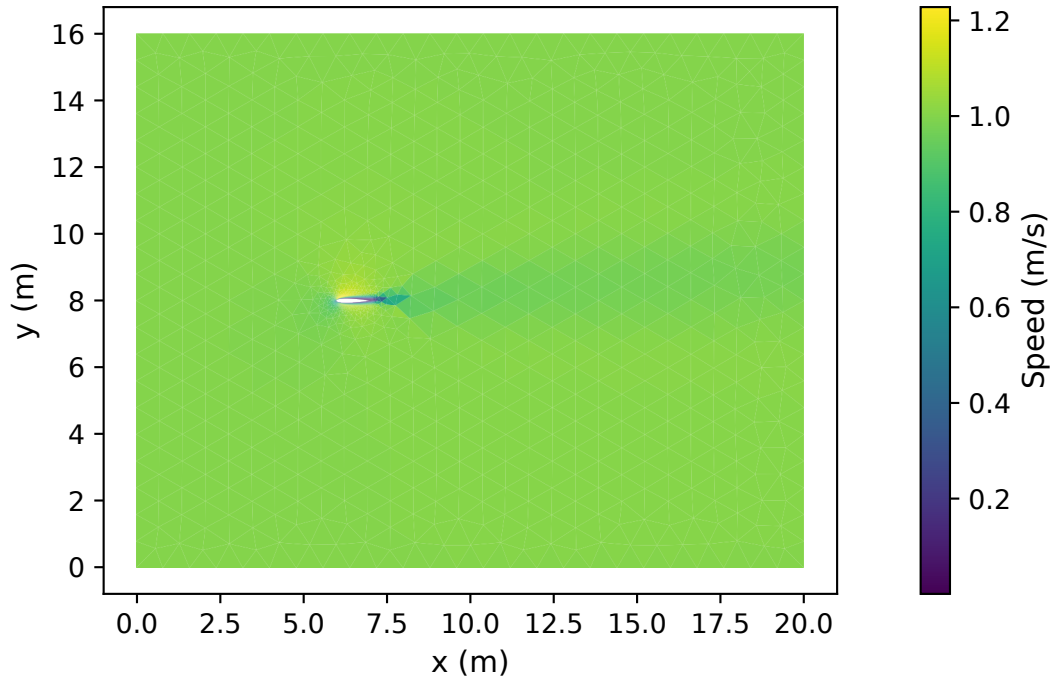
naca: 3934 cells



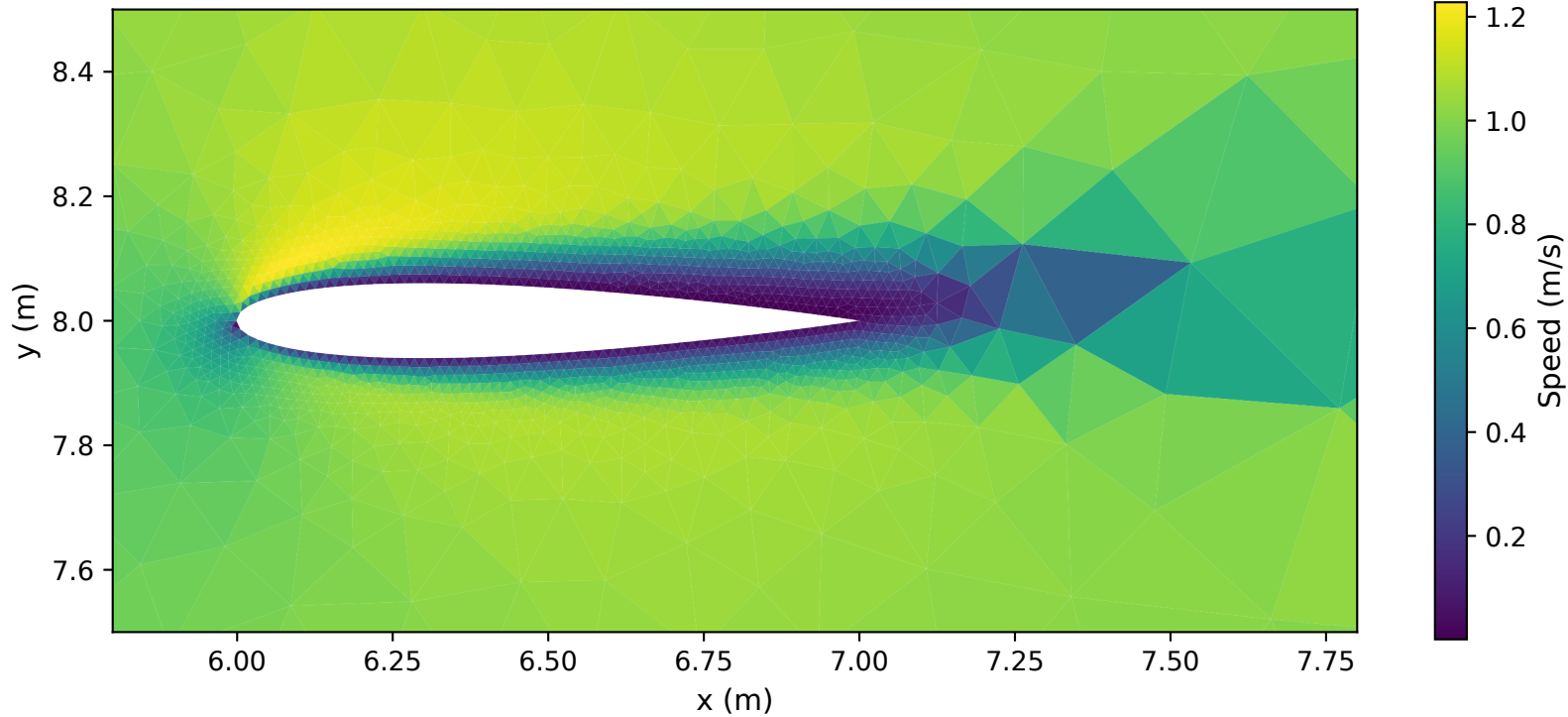
naca: 3934 cells



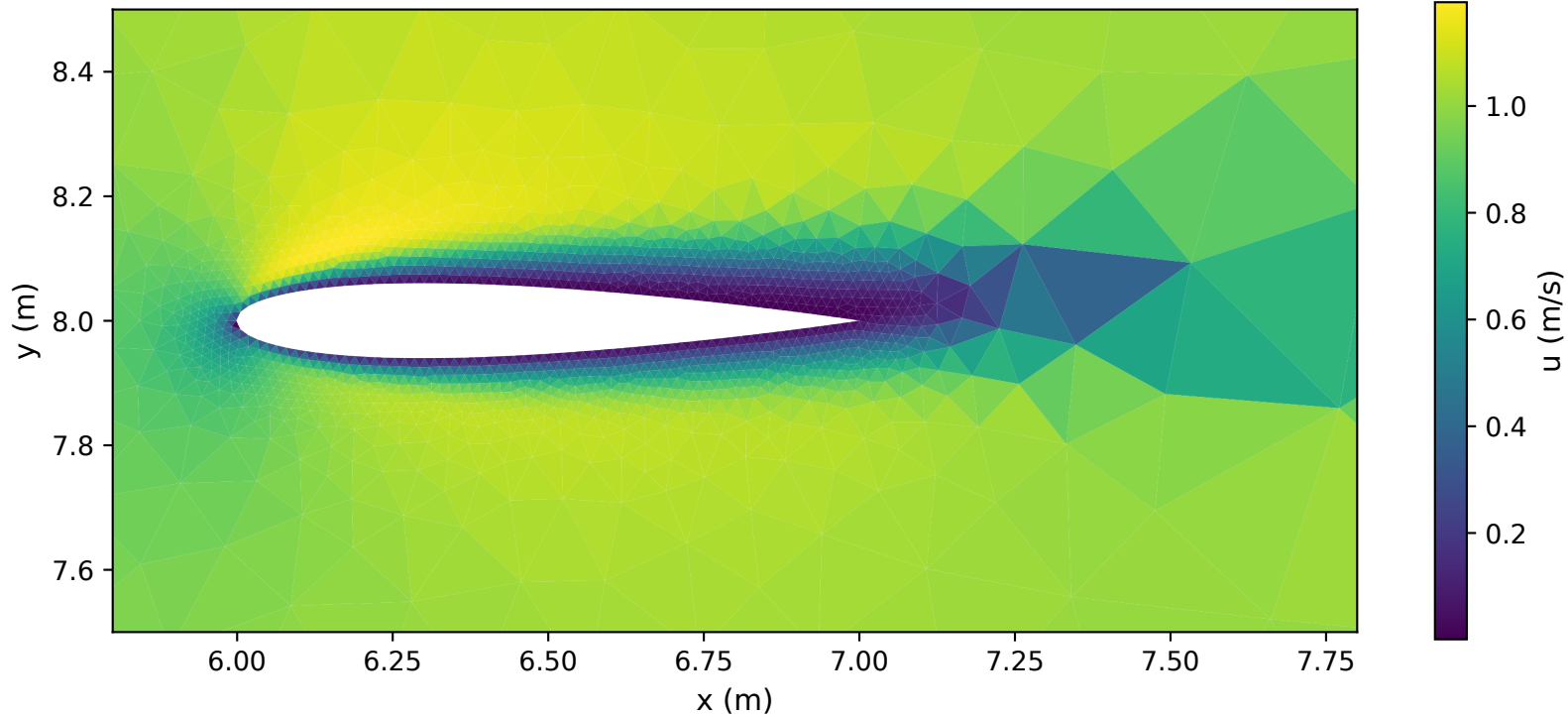
naca: 3934 cells



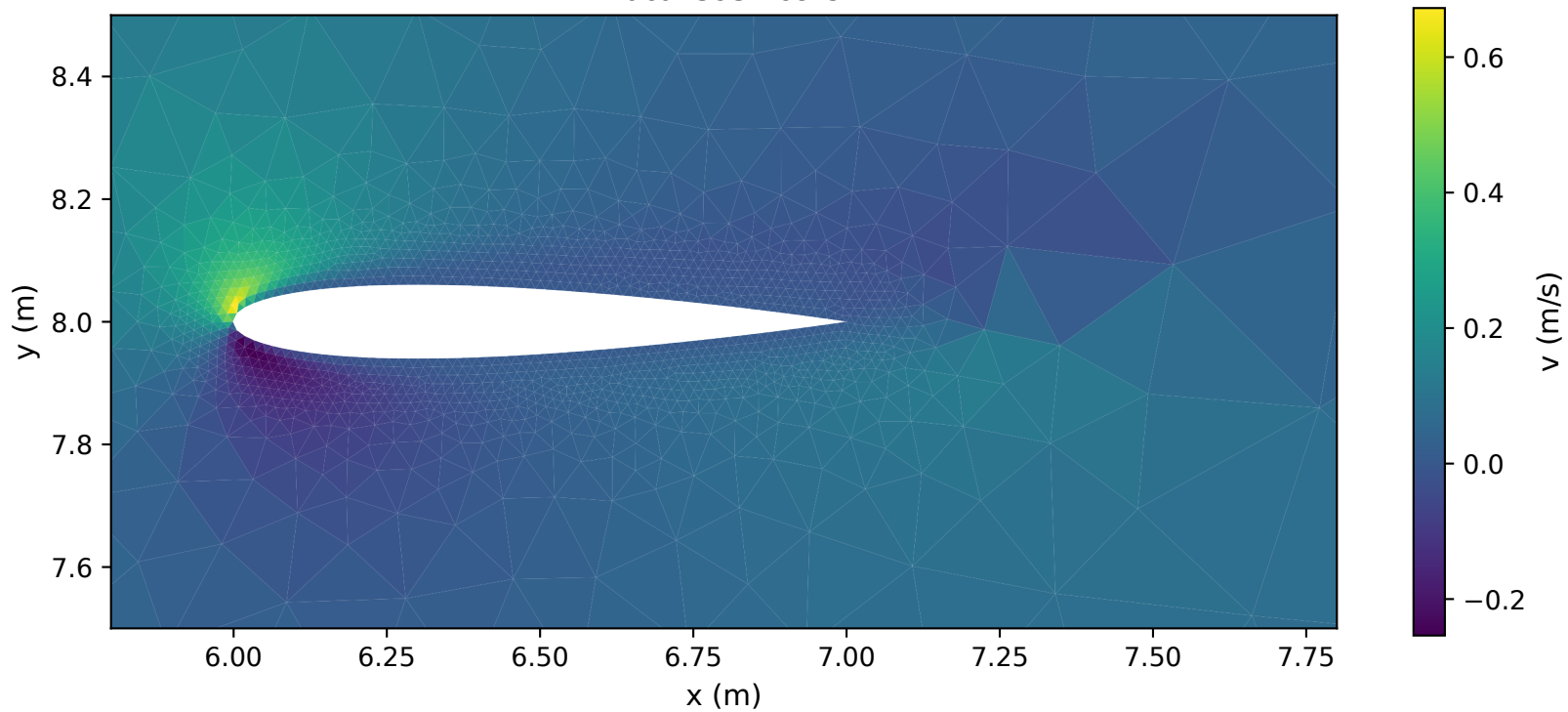
naca: 3934 cells



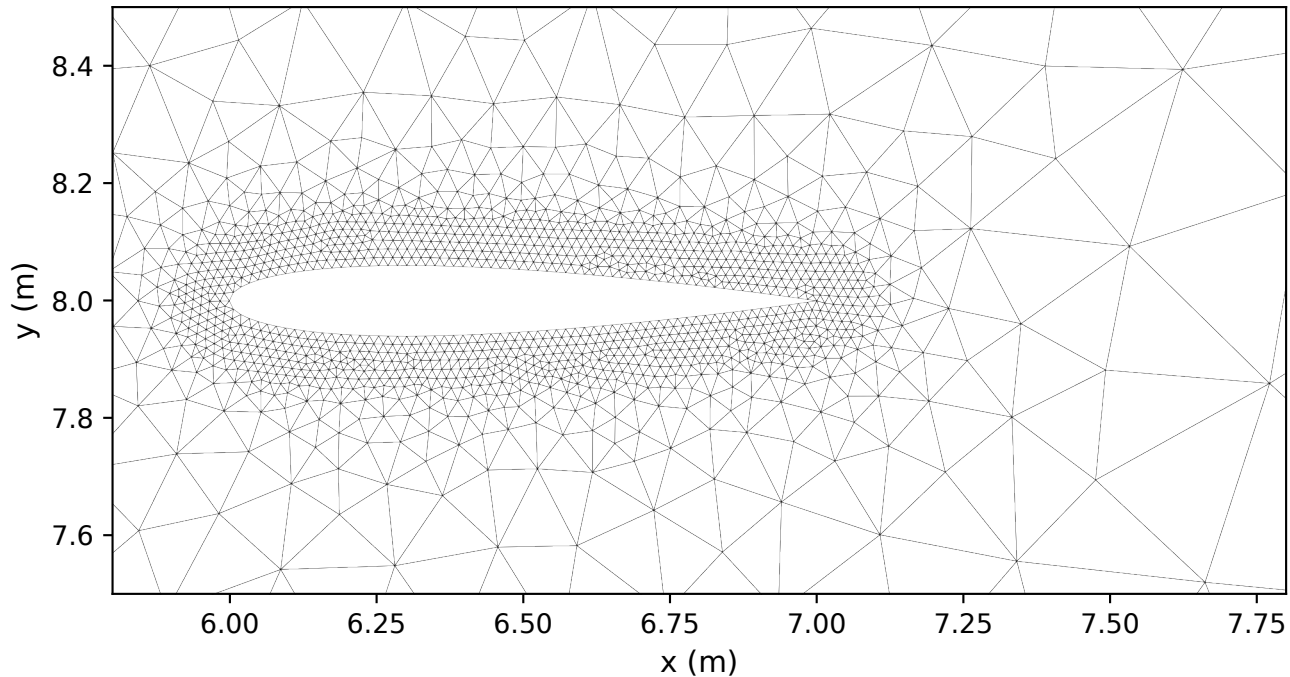
naca: 3934 cells

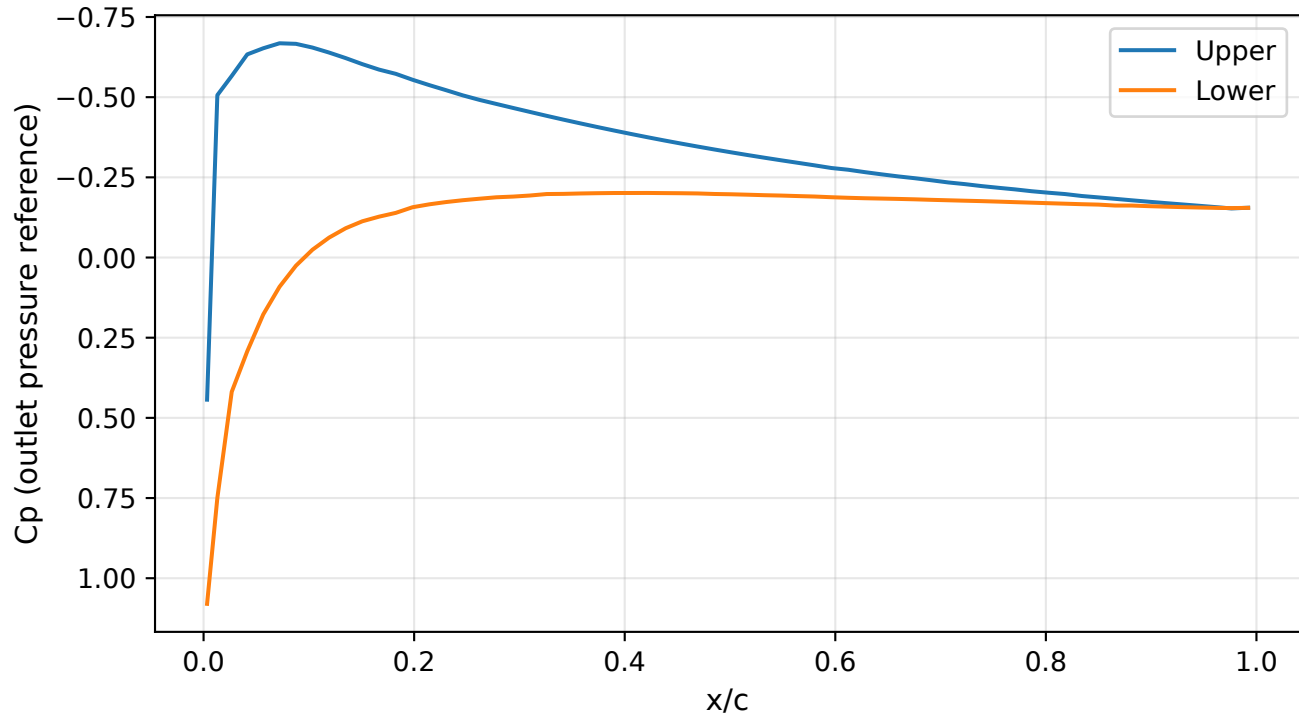


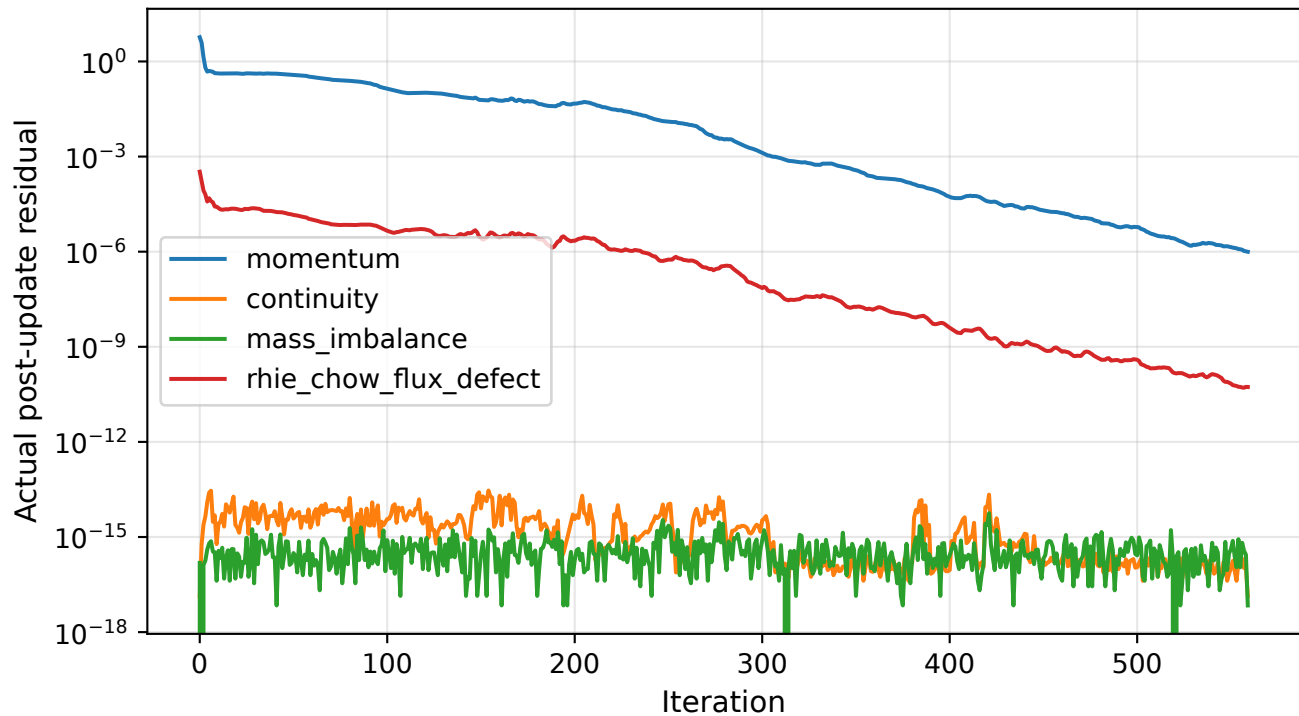
naca: 3934 cells



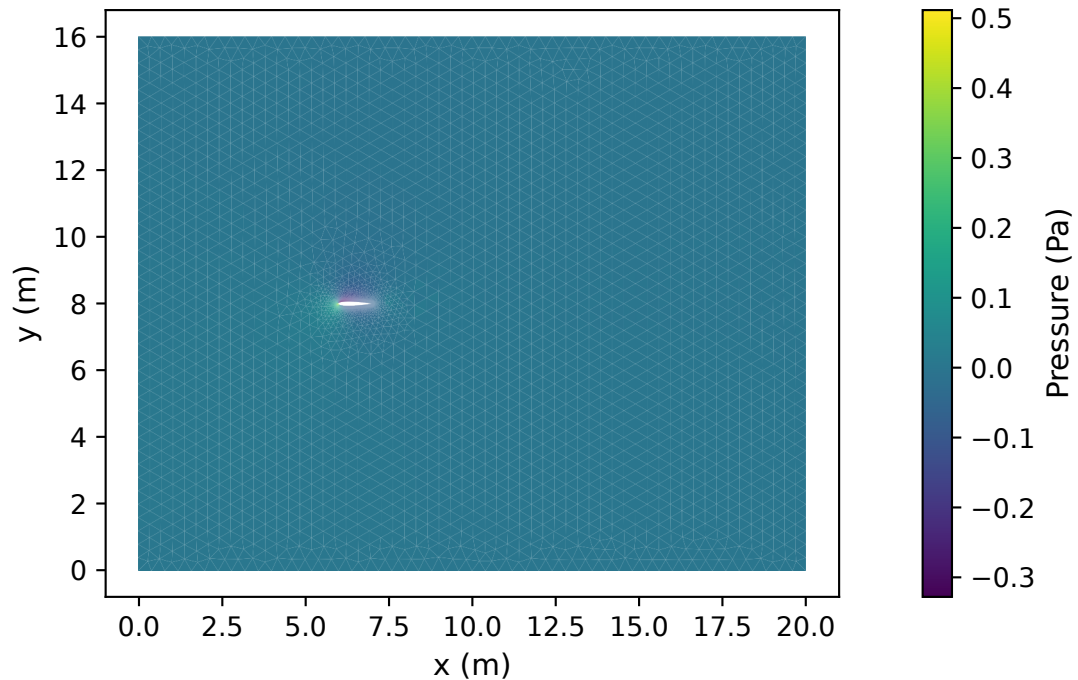
Actual body-fitted mesh



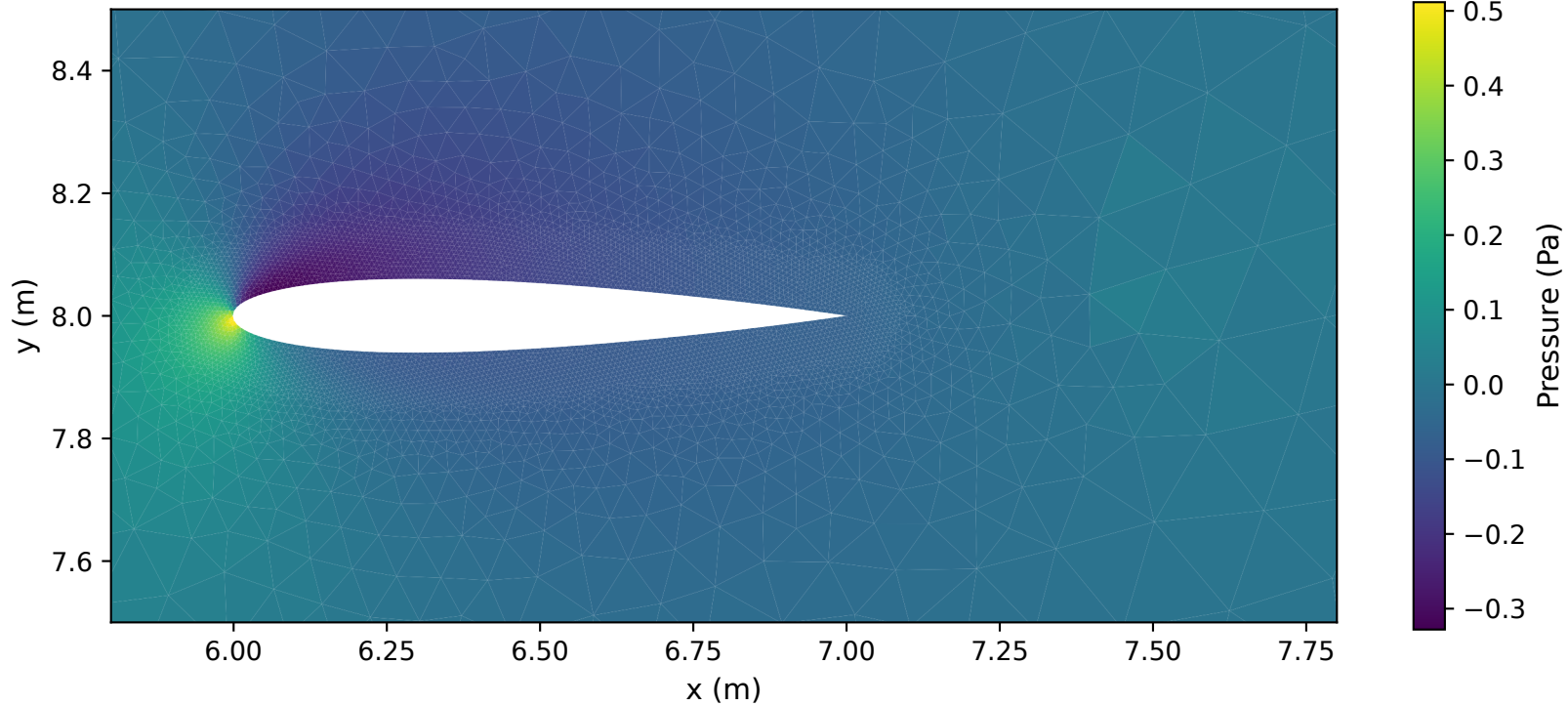




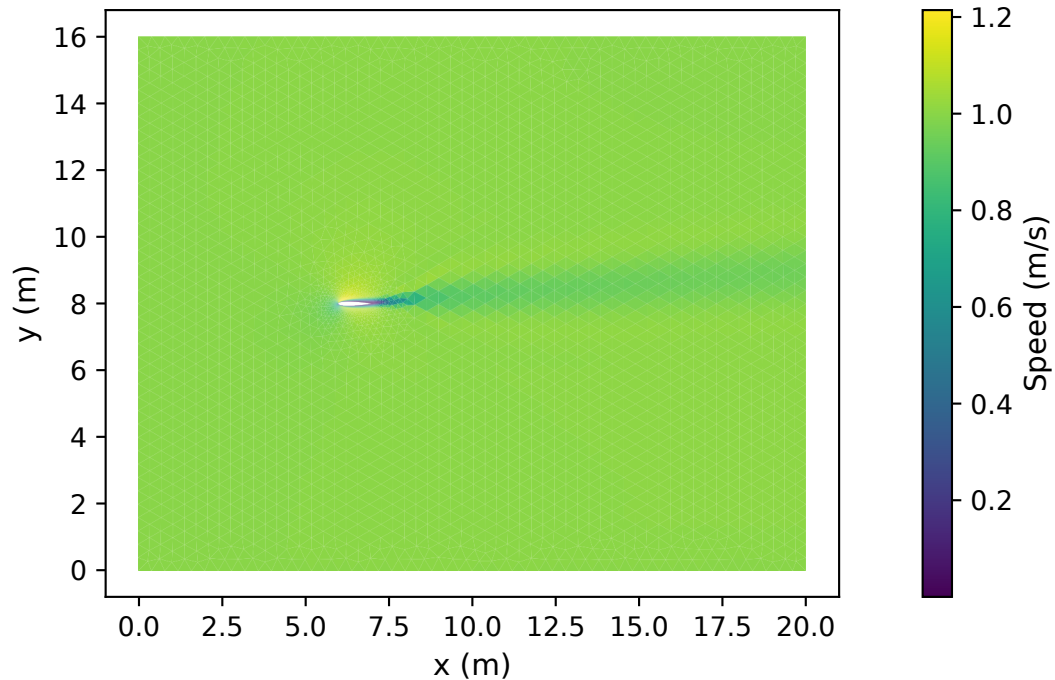
naca: 15018 cells



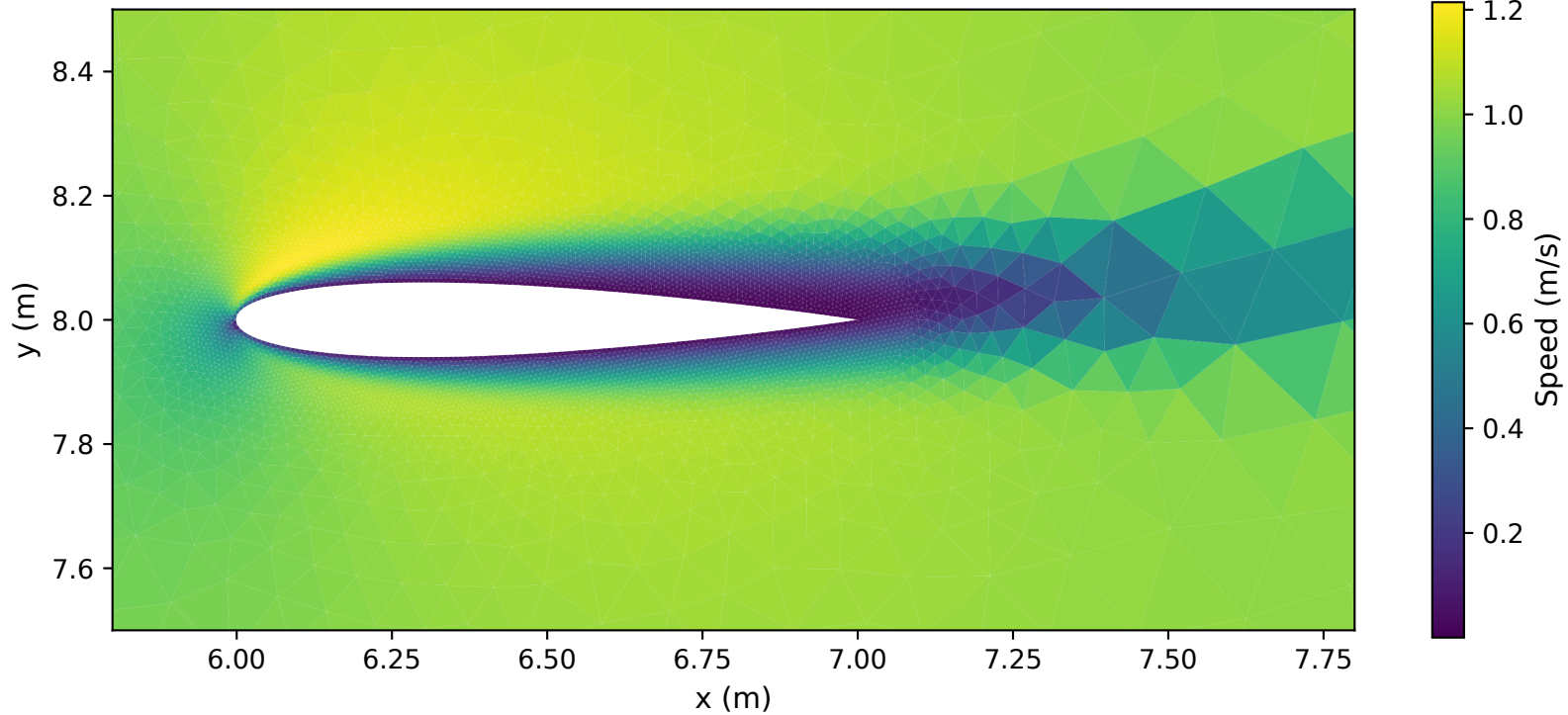
naca: 15018 cells



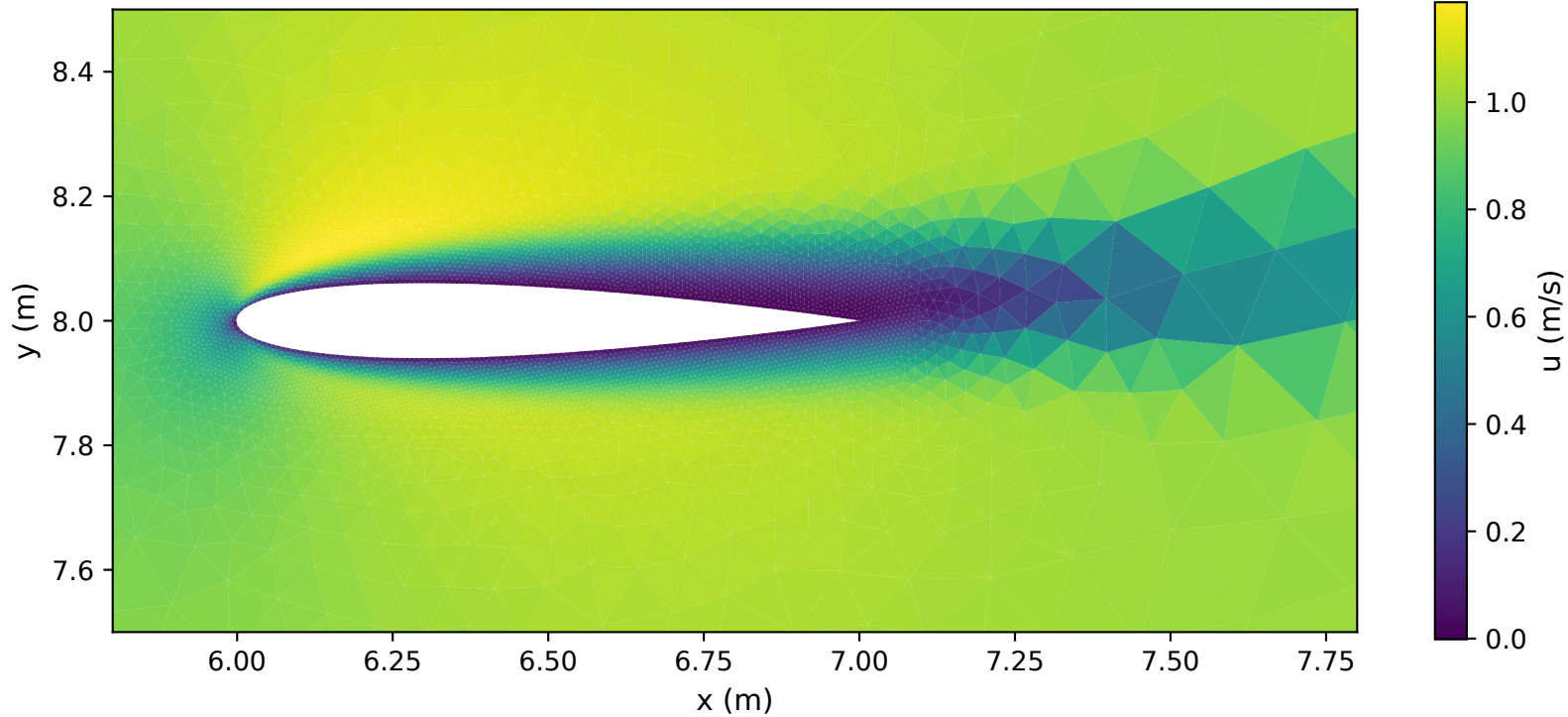
naca: 15018 cells



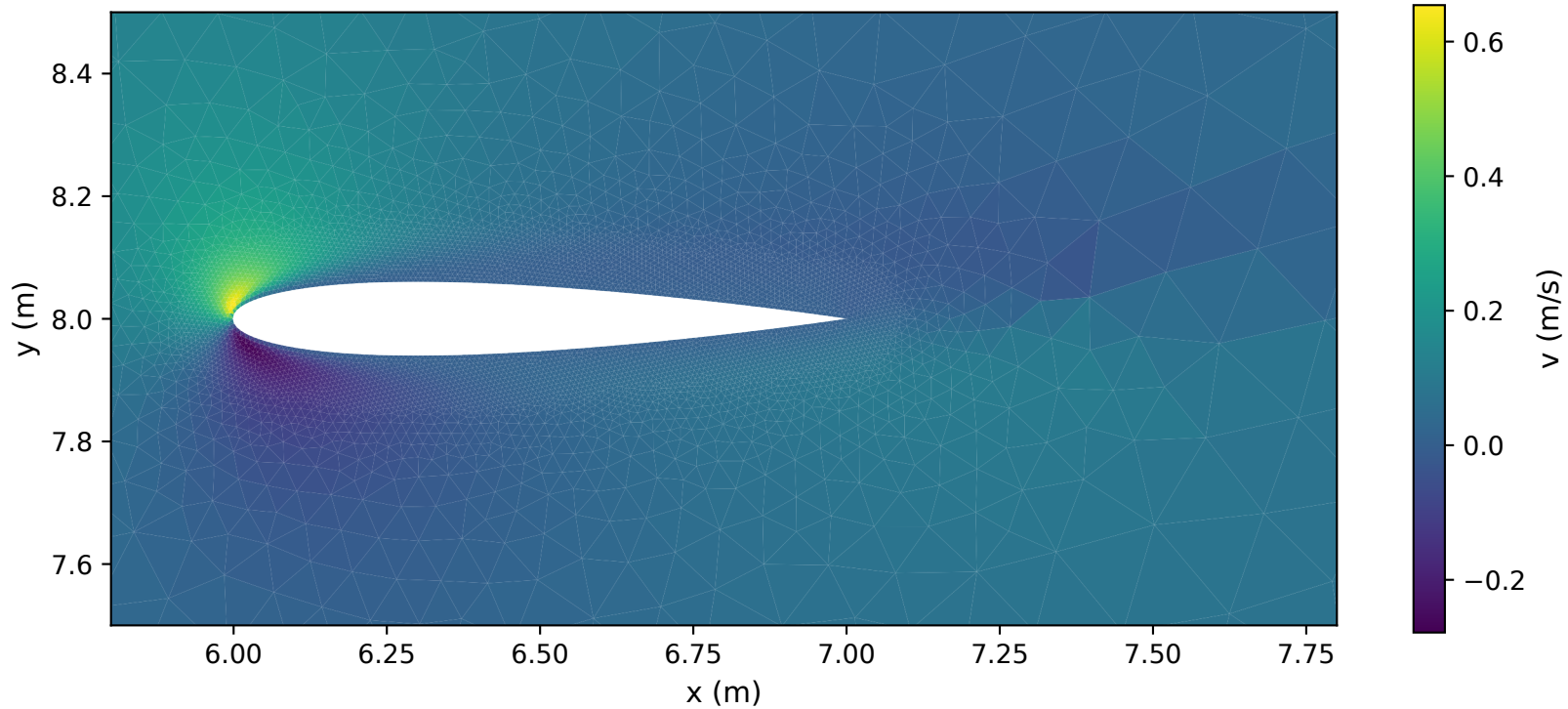
naca: 15018 cells



naca: 15018 cells



naca: 15018 cells



Actual body-fitted mesh

